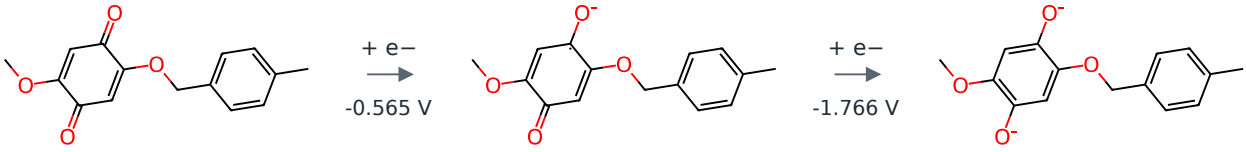
Computed capped-model charge/spin ladders

1 Methoxy BQ

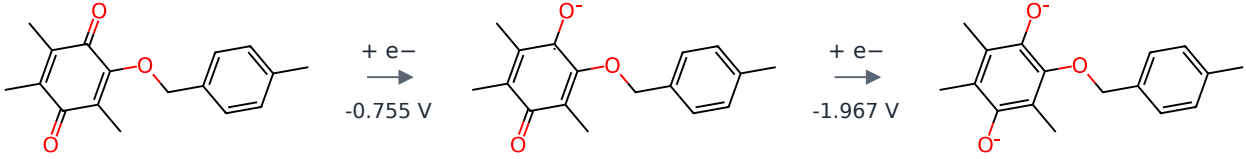


Neutral · q = 0
M = 1 (selected model)

Radical anion · q = -1
M = 2 (selected model)

Dianion · q = -2
M = 1 (selected model)

2 Trimethyl BQ

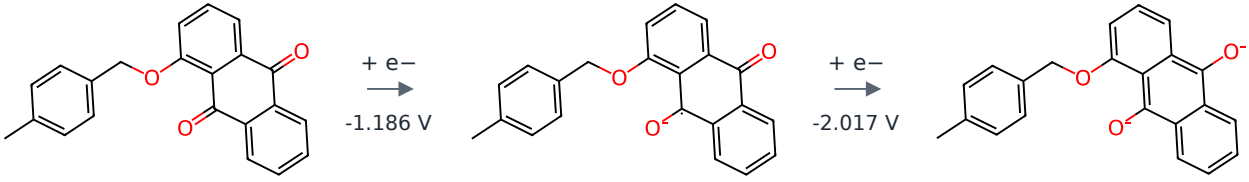


Neutral · q = 0
M = 1 (selected model)

Radical anion · q = -1
M = 2 (selected model)

Dianion · q = -2
M = 1 (selected model)

3 1-O AQ



Neutral · q = 0
M = 1 (selected model)

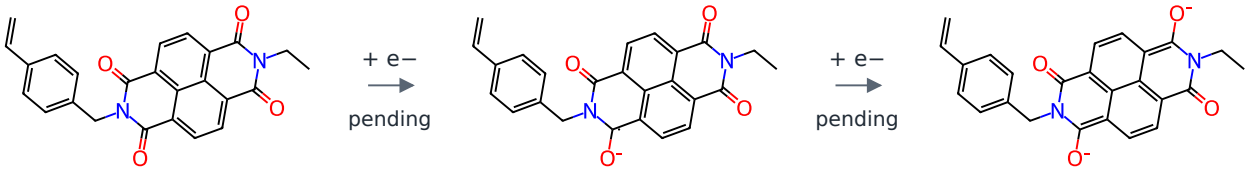
Radical anion · q = -1
M = 2 (selected model)

Dianion · q = -2
M = 1 (selected model)

Illustrative resonance forms; charge/spin may be delocalized. No proton transfer is implied.
Potentials: calculated MeCN vs Fc/Fc+. Arrows show electron count, not demonstrated reversibility.

Proposed vinyl-monomer ladders · uncomputed

P1 NDI vinyl monomer

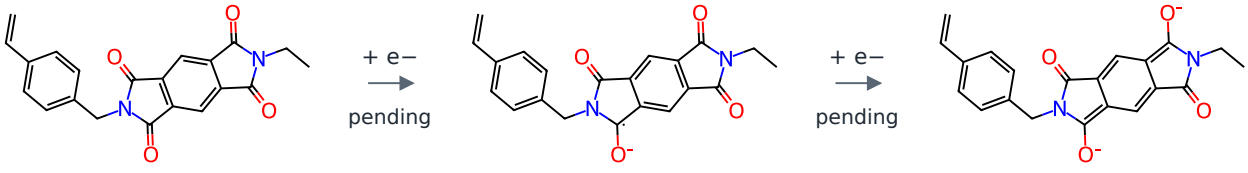


Neutral · q = 0
M = 1 (proposed)

Radical anion · q = -1
M = 2 (proposed)

Dianion · q = -2
M = 1 (proposed)

P2 PMDI vinyl monomer

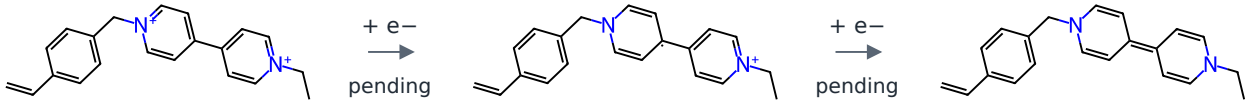


Neutral · q = 0
M = 1 (proposed)

Radical anion · q = -1
M = 2 (proposed)

Dianion · q = -2
M = 1 (proposed)

P3 Viologen vinyl monomer



Dication · q = +2
M = 1 (proposed)

Radical cation · q = +1
M = 2 (proposed)

Neutral · q = 0
M = 1 (proposed)

Illustrative resonance forms; charge/spin may be delocalized. No proton transfer is implied.
Vinyl-state properties and polymer performance are uncomputed. M is the proposed multiplicity.